MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYOLEFIN, SEMI-RIGID, CROSSLINKED

This specification is approved for use by all Department and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

REQUIREMENTS:

Continuous operating temperature range: -55°C (-67°F) to +135°C (+275°F).

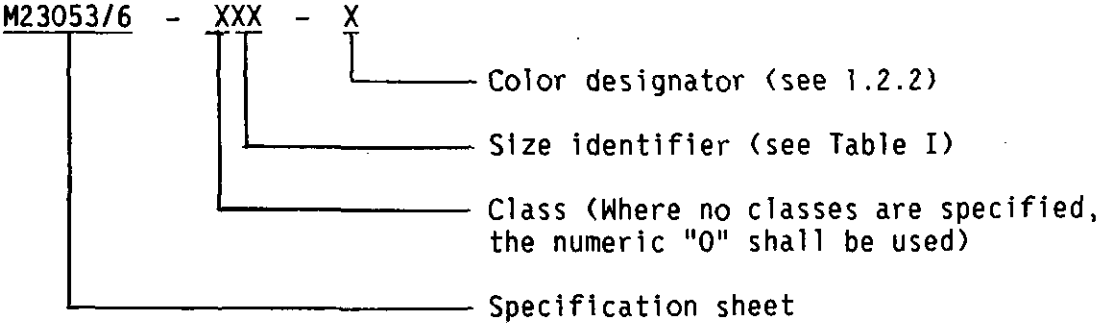
Classification: The heat shrinkable sleeving shall be furnished in the following classes, as specified (see 6.2.1):

Class 1 - Flame resistant
Class 2 - Flammable (clear only)

Color: Class 2 shall be furnished in clear. Class 1 shall be furnished in colors only. Unless otherwise specified by the acquisition activity (see 6.2.1), each size of Class 1 sleeving shall be furnished in the color designated in Table I. All colors shall conform to Class 1 of MIL-STD-104.

Longitudinal change: ± 5 percent.

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



NOTE: Entire specification sheet revised.

AMSC N/A FSC 5970

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Example: Class 2, 0.125 inch (3.18 mm) as supplied ID sleeving shall be identified as M23053/6-204-C.

TABLE I. Construction details, inches (mm). 1/

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/6-101-4	.046(1.17)	.023(.58)	.020 ± .003(.508 ± .076)
M23053/6-102-9	.063(1.60)	.031(.79)	.020 ± .003(.508 ± .076)
M23053/6-103-6	.093(2.36)	.046(1.18)	.020 ± .003(.508 ± .076)
M23053/6-104-2	.125(3.18)	.062(1.59)	.020 ± .003(.508 ± .076)
M23053/6-105-5	.187(4.75)	.093(2.36)	.025 ± .003(.635 ± .076)
M23053/6-106-8	.250(6.35)	.125(3.18)	.025 ± .003(.635 ± .076)
M23053/6-107-1	.375(9.53)	.187(4.75)	.030 ± .005(.762 ± .127)
M23053/6-108-9	.500(12.7)	.250(6.35)	.030 ± .005(.762 ± .127)
Class 2			
M23053/6-201-C	.046(1.17)	.023(.58)	.020 ± .003(.508 ± .076)
M23053/6-202-C	.063(1.60)	.031(.79)	.020 ± .003(.508 ± .076)
M23053/6-203-C	.093(2.36)	.046(1.17)	.020 ± .003(.508 ± .076)
M23053/6-204-C	.125(3.18)	.062(1.58)	.020 ± .003(.508 ± .076)
M23053/6-205-C	.187(4.75)	.093(2.36)	.025 ± .003(.635 ± .076)
M23053/6-206-C	.250(6.35)	.125(3.18)	.025 ± .003(.635 ± .076)
M23053/6-207-C	.375(9.53)	.187(4.75)	.030 ± .005(.762 ± .127)
M23053/6-208-C	.500(12.7)	.250(6.35)	.030 ± .005(.762 ± .127)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data

2/ Wall thickness values are less when shrinkage is restricted.

3/ The color codes cited above for Class 1 sleevings are preferred acquisition colors. Class 1 sleevings are also available in other colors (see 1.2.2)

Unrestricted shrinkage: Test method 4.6.5, 200° ± 2°C (392° ± 4°F) for 3 minutes.

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TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 250° ± 3°C (482° ± 6°F)
Secant modulus, psi (MPa), minimum	25,000 (172.4)	4.6.13.1, 2 percent strain, ASTM D 882
Color stability	Pass	4.6.16 175° ± 2°C (347° ± 4°F), 24 hours
Concentricity	70% minimum	4.6.3.3
Clarity stability, Class 2	Pass	4.6.17, 175° ± 2°C (347° ± 4°F), 24 hours
Restricted shrinkage	No cracks	4.6.6.1.1 175° ± 2°C (347° ± 4°F)
Voltage withstand	Pass	4.6.6.2
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, inches/minute psi (MPa), minimum	2,000(13.8)	4.6.14 ASTM D 638, 2
Ultimate elongation, inches/minute percent, minimum	200	4.6.14 ASTM D 638, 2
Dielectric strength, volts/mil (Kv/mm), minimum	500 (19.7)	4.6.2 ASTM D 2671
Volume resistivity, Ohm-cm, minimum	Class 1 - 10^{14} Class 2 - 10^{16}	4.6.2. ASTM D 876
Dielectric constant, maximum 3/	Class 1 - 3.2; Class 2 - 2.7	4.6.2 ASTM D 150

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Specific gravity, maximum	Class 1 - 1.35; Class 2 - 1.00	4.6.2 ASTM D 792
Water absorption, percent, maximum	Class 1 - 0.5; Class 2 - 0.2	4.6.2 ASTM D 570, 24 hrs at 23°C
Corrosion	No corrosion	4.6.10.1 and 4.5.10.2, 150° ± 2°C (302° ± 4°F), 16 hours
Low temperature flexibility	No cracking	4.6.7.1 -55° ± 1°C (-67° + 2°F)
Heat resistance, properties after:		4.6.9 175° ± 2°C (347° ± 4°F) for 168 hours
Ultimate elongation, percent, minimum	100	
Fluid resistance, properties after:		4.6.11
Tensile strength, psi (MPa), minimum	1,600(11.1)	
Dielectric strength, volts/mil (Kv/mm), minimum	400 (15.8)	
Flammability	Class 1 only - self-extinguishing within one minute and no more than 25 percent of indicator flag burned or charred. No dripping or flowing of large sizes; Class 2 - N/A	4.6.15.1
Fungus resistance 2/	No Growth	4.6.2 ASTM G 21

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TABLE II. Physical properties. 1/ - Continued

1/ Unless otherwise specified, the requirements, test procedures and conditions are for all classes.

2/ MIL-I-23053/6 materials do not normally support fungus growth. Performance of this requirement is at the option of the acquiring activity (see 6.2.1).

3/ Dielectric constant is a requirement only when specified in acquisition document.

Shelf life conditions: Supplier shall certify to storage at 40°C (104°F) for four years. Conformance to 3.5. See 3.5.2 for shelf life extension.

NOTE:

a. Flammability - Class 2 heat shrinkable sleeving described in this specificaton sheet have not been flame retarded and will burn readily. These sleevings (Class 2) shall not be considered for acquisition when flame resistance is an essential requirement.

Intended use: Heat shrinkable semi-rigid polyolefin sleeving is used in wire identification by marking or strain relief applications.

Cross reference: Classes 1 and 2 of MIL-I-23053/6C are equivalent to Classes 1 and 2 respectively, of MIL-I-23053/6B. Class 3 has been deleted with Class 1 sleevings as a suitable substitute.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - EA, MI
Navy - EC
Air Force - 99
DLA - GS

User activities:
Army - ME
Navy - MC

Preparing activity:
Navy - AS

(Project 5970-0577-06)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-23053D/6C	INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABL GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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